SELECTIVE NOISE CANCELLATION USING MACHINE LEARNING

A Mini Project Report

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Bachelor of Technology

in

Computer Science and Engineering

by

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CERTIFICATE

This is to certify that the report entitled **Selective Noise Cancellation Using Machine Learning** submitted by **Suhair K** (NSS22EC118), **Swathy Krishna K** (NSS22EC119), & **Vismaya J** (NSS22EC128) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. Minor degree in Computer Science and Engineering is a bonafide record of the Mini Project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the Minor project report **Selective Noise Cancellation using Machine learning**, submitted for partial fulfillment of the requirements for the award of Minor degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Prof.Kiran V K.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

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Abstract

In real-world environments, speech signals are often corrupted by background noise, leading to poor intelligibility and degraded performance in communication systems and speech recognition applications. This project proposes a deep learning-based approach for real-time speech noise cancellation, aiming to enhance the clarity and quality of speech in noisy conditions. A denoising autoencoder model is developed and trained using paired clean and noisy audio datasets.The model learns to map noisy spectrogram features to their corresponding clean versions. The processed spectrogram is then converted back into an audio waveform to produce enhanced speech. The system utilizes Python libraries such as Librosa and PyTorch for audio processing and model training.This project demonstrates the effectiveness of deep learning techniques in speech signal enhancement and has potential applications in voice communication, hearing aids, and speech-controlled systems.

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Chapter 1

Introduction

1.1 Background

Noise pollution has become an inevitable part of daily life, especially in cities, industries, and public spaces. Continuous exposure to unwanted noise can lead to stress, reduced productivity, and even long-term hearing damage. To minimize these effects, Active Noise Cancellation (ANC) technology has been developed, which works by generating an anti-phase sound wave to cancel incoming noise. Although this method effectively reduces background disturbances, it does not differentiate between useful and irrelevant sounds. For instance, traditional ANC systems may suppress both a car horn and engine noise equally, which can lead to safety concerns. Recent advancements in machine learning and signal processing offer the potential to overcome this limitation by enabling systems to identify and treat sounds based on their importance and context. [1]

1.2 Motivation

In modern urban environments, people are constantly surrounded by various types of noise that can affect concentration, communication, and hearing health. Conventional Active Noise Cancellation (ANC) systems eliminate all sounds without distinction, often suppressing important cues such as sirens, car horns, or alarms that are vital for safety. To overcome this limitation, there is a growing need for an intelligent system capable of selectively filtering unwanted noise while preserving essential sounds. By leveraging machine learning techniques, this project aims to develop a selective noise cancellation system that enhances listening comfort, safety, and clarity in real-world acoustic environments.

1.3 Objectives

The primary objectives of this project are outlined below:

(i) *To develop a machine learning–based selective noise cancellation system:* The project aims to design and implement a system that uses machine learning algorithms to intelligently identify and filter unwanted noise from audio signals.

(ii) *To separate primary audio from environmental noise:* The system will distinguish between the main audio content (such as speech or music) and background sounds using source separation techniques like Demucs.

(iii) *To classify sounds as important or non-important:* A classification model, trained on labeled audio data, will categorize sounds into critical (e.g., sirens, horns, alarms) and non-critical (e.g., chatter, engine hum, drilling) categories.

(iv) *To suppress unwanted sounds while allowing critical alerts to pass:* Based on the classification results, the model will selectively eliminate non-important noise while preserving important sounds that ensure safety and awareness.

(v) *To reconstruct a clear and balanced audio output in real time:* The final output signal will be reconstructed by combining the preserved audio components, providing clarity, comfort, and situational awareness to the listener.

1.4 Challenges in Selective Noise Cancellation

Developing an intelligent, selective noise cancellation system presents several technical and practical challenges.

(i) *Accurate Sound Classification:*Distinguishing between essential sounds like sirens or alarms and background noise is difficult because different sounds often overlap in real-world environments.

(ii) *Real-Time Processing:*The system must classify and filter sounds quickly without delay, which requires efficient algorithms and optimized processing.

(iii) *Data Quality and Diversity:*Reliable classification depends on high-quality and varied audio data. Limited or unbalanced datasets can reduce the system's accuracy.

(iv) *Hardware and Performance Optimization:*Deploying the system on low-power devices like headphones or smartphones requires reducing computational load while maintaining clarity and performance..

1.5 Proposed Solution

To overcome the limitations of traditional noise cancellation, this project proposes an intelligent, machine learning–based selective noise cancellation system. The solution focuses on allowing important sounds to pass through while suppressing unwanted background noise.

(i) *Source Separation Using Deep Learning:*The incoming audio signal is separated into main audio and environmental noise using advanced source separation models. This allows the system to process different sound components independently.

(ii) *Sound Classification:*A trained machine learning model classifies environmental sounds as important (e.g., sirens, horns, alarms) or non-important (e.g., traffic rumble, chatter, drilling).

(iii) *Selective Filtering:*Only non-important noise is suppressed, while critical sounds and the main audio are preserved. This ensures safety and situational awareness without compromising listening comfort.

(iv) *Real-Time Implementation:*The entire process is designed to work in real time and can be deployed on practical devices such as headphones, smartphones, and vehicles.

Chapter 2

Literature Review

Noise cancellation has been an active area of research for several decades, with early work focusing on traditional active noise control (ANC) techniques such as the filtered-x least mean square (FxLMS) algorithm. [2] These methods generate an anti-noise signal to cancel the unwanted sound at the error microphone, and they are widely appreciated for their simplicity and low computational cost. However, linear ANC systems face several challenges in practical scenarios, especially in the presence of non-linear noise sources, dynamic environments, and multiple interfering signals. While non-linear extensions have been proposed, they often increase the complexity of implementation and reduce adaptability in real-time situations. As a result, traditional approaches are not always effective in modern noisy environments such as urban roads, transportation hubs, or industrial settings.

Recent research has shifted toward deep learning–based approaches to overcome these limitations. In particular, [3] introduces a framework where ANC is formulated as a supervised learning problem, allowing neural networks to learn the complex relationship between reference and noise signals. Deep models such as convolutional recurrent networks (CRN) provide better adaptability to non-stationary noise and can handle real-time signal processing more effectively. Other studies [1] have also explored neural network–based ANC systems that improve robustness and accuracy in practical applications. These methods make use of advanced feature extraction techniques, including the short-time Fourier transform (STFT) and MFCCs, which help the model distinguish between different sound sources. Unlike traditional ANC, deep learning–based systems can be designed for selective noise cancellation, where important sounds like sirens, alarms, and horns are preserved while suppressing irrelevant background noise. This makes them more suitable for real-world scenarios where both clarity and safety are essential.

Chapter 3

Feasibility Analysis

This chapter evaluates the feasibility of implementing the proposed **Selective Noise Cancellation Using Machine Learning** framework. The feasibility analysis examines key aspects including data availability, technical requirements, computational resources, and anticipated challenges to ensure that the project objectives are achievable within the given scope and constraints.

3.1 Data Availability

(i) **Dataset Characteristics:** The project uses the *UrbanSound8K* dataset, which consists of 8,732 labeled audio clips representing real-world urban environments. The dataset includes critical alert sounds such as sirens and car horns, as well as background noise like traffic, drilling, and chatter. This makes it suitable for training a model that distinguishes between essential and non-essential sounds.

(ii) **Existing Datasets:** UrbanSound8K is freely available and well-annotated. Its standardized format (44.1 kHz, 16-bit WAV) simplifies preprocessing and model training. Additional audio samples can be collected to further improve model robustness if needed.

(iii) **Data Quality and Preprocessing:** The dataset can be preprocessed efficiently using existing audio processing tools to extract features such as MFCCs, pitch, and amplitude envelope. These features support accurate classification during model training. The availability of a structured and high-quality dataset ensures smooth data handling throughout the project.

3.2 Technical Feasibility

(i) **Pre-trained Models and Frameworks:** The project leverages `Demucs` for source separation and a CNN-based classifier for sound categorization. These models are well-

documented and proven in audio processing tasks, making them reliable components for the system.

(ii) **Software Libraries:** Implementation will be carried out in `Python`, using deep learning frameworks like `TensorFlow` and `PyTorch`, and audio libraries such as `librosa` and SoundFile. These tools provide built-in functions for signal processing, model building, and evaluation, enabling a modular and maintainable architecture.

(iii) **Deployment Feasibility:** The solution can be deployed on headphones, smartphones, or vehicles after optimization. Since all frameworks are open-source and lightweight models can be used, the technical implementation is practical and cost-effective.

3.3 Computational Feasibility

(i) **Hardware Requirements:** Model training and evaluation can be performed using cloud services such as Google Colab (GPU enabled), which eliminates the need for expensive local hardware during development.

(ii) **Training Time and Resources:** Given the moderate size of UrbanSound8K and the efficiency of CNN and Demucs models, training and testing can be completed within the project timeline. Cloud GPU support ensures faster model convergence and iteration.

(iii) **Scalability:** The system architecture allows scaling to larger datasets or real-time applications with minor modifications. After training, the model can be compressed and deployed on edge devices with minimal computational overhead.

3.4 Anticipated Challenges

(i) **Overlapping Sound Sources:** Urban noise often contains multiple sounds occurring simultaneously, which may make it difficult for the model to isolate and classify signals accurately.

(ii) **Real-Time Processing:** Maintaining low latency during live noise filtering is challenging. The system must process signals quickly without compromising accuracy or clarity.

(iii) **Device-Level Optimization:** Deploying on resource-limited devices requires careful optimization techniques such as model quantization, pruning, and efficient signal-processing pipelines to maintain acceptable performance under hardware constraints.

Chapter 4

Methodology

This chapter details the proposed approach for implementing selective noise cancellation using machine learning. The methodology follows a systematic process beginning with data collection and preprocessing of audio signals that contain both primary and environmental sounds. Using the Demucs model for source separation, the system splits the input audio into main content (such as speech or music) and background noise. The extracted noise components are then analyzed by a machine learning classifier to identify important and non-important sounds. Based on this classification, the system selectively suppresses unwanted noise while preserving essential sounds. Finally, the processed signals are reconstructed to produce a clear, balanced audio output in real time.

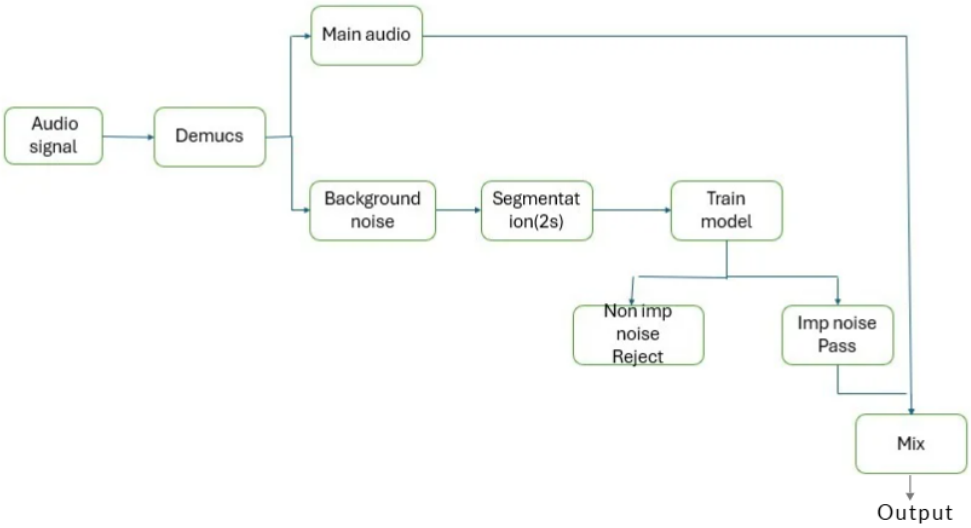


Figure 4.1: Overall system architecture and workflow

4.1 System Architecture

The proposed system architecture is designed to intelligently separate and classify audio signals in real time. It consists of the following major components:

- **Input Acquisition:** Capturing audio containing both foreground signals (speech, music, alerts) and background noise (traffic, chatter, drilling).
- **Source Separation:** Using a deep learning-based separation model to split the input into main audio and background noise.
- **Feature Extraction and Classification:** Identifying important versus non-important noise using a CNN classifier.
- **Signal Reconstruction:** Combining the main audio with the important noise while suppressing non-important noise.
- **Output:** Generating the enhanced, clear audio signal suitable for real-time use.

4.2 Audio Preprocessing

Before training and classification, the input audio undergoes several preprocessing steps to ensure quality and consistency:

1. **Resampling:** All audio is converted to a consistent 44.1 kHz sample rate.
2. **Normalization:** Volume levels are adjusted to maintain uniform amplitude.
3. **Framing and Windowing:** Audio signals are divided into short overlapping frames to extract fine-grained temporal information.
4. **Noise Reduction:** Basic denoising is applied to reduce unwanted artifacts before model processing.

4.3 Source Separation

The core of the system involves separating the input audio signal into two distinct components:

- **Foreground (Main Audio):** Contains speech, music, or primary signal.
- **Background Noise:** Contains environmental sounds such as sirens, chatter, traffic, or drilling.

This step is implemented using `Demucs`, a pre-trained deep learning model for audio source separation. It allows independent processing of background noise for further classification.

4.4 Feature Extraction

Feature extraction is essential to provide the classifier with meaningful audio representations. The following features are used:

- Mel-Frequency Cepstral Coefficients (MFCCs)
- Zero Crossing Rate (ZCR)
- Spectral Centroid and Bandwidth
- Pitch and Amplitude Envelope

These features help the classifier distinguish between different types of background sounds with high accuracy.

4.5 Sound Classification

The background noise obtained from source separation is passed to a Convolutional Neural Network (CNN) classifier trained on the UrbanSound8K dataset. The classifier:

- Identifies and labels the background sound as **important** (e.g., sirens, car horns) or **non-important** (e.g., chatter, drilling).
- Uses these labels to decide which sounds should be preserved and which should be suppressed.

4.6 Sound Reconstruction

After classification, the system reconstructs the output audio:

1. Non-important noise is suppressed.
2. Important noise is retained and combined with the foreground signal.
3. The final signal is normalized and exported as the enhanced output.

4.7 Summary

This chapter outlined the methodology used to build the proposed system. Starting from audio acquisition, the signal is preprocessed, separated, classified, and selectively filtered to produce a clean output signal. The use of deep learning models enables accurate detection of critical alert sounds while ensuring real-time performance suitable for practical applications.

Chapter 5

Tools and Frameworks Used

This chapter provides an overview of the tools, libraries, and frameworks used in developing the proposed intelligent audio filtering and classification framework. Each tool played a significant role in different stages of the workflow, such as audio preprocessing, source separation, feature extraction, model training, visualization, and deployment.

5.1 Programming Languages and Environments

1. **Python:** The primary programming language used for implementing the proposed audio filtering and classification model. Python offers extensive support for machine learning, deep learning, and audio processing through a wide range of open-source libraries.

2. **Jupyter Notebook and Google Colab:** Interactive development environments used for coding, debugging, and running experiments. Google Colab provides GPU acceleration for faster model training and testing.

5.2 Deep Learning Frameworks

1. **PyTorch:** The main deep learning framework used for implementing and training the Convolutional Neural Network (CNN) classifier. PyTorch supports dynamic computation graphs and efficient GPU utilization, making it suitable for model experimentation and fine-tuning.

2. **TensorFlow:** Used during early experimentation for comparing neural network architectures and validating model performance across different frameworks.

5.3 Audio Processing and Feature Extraction Tools

1. **Librosa:** A specialized Python library for audio signal analysis. It was used for feature extraction tasks such as computing Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, and zero-crossing rate.

2. **Demucs:** A state-of-the-art deep learning model used for source separation. It divides mixed audio into foreground and background components, enabling focused analysis of environmental sounds.

5.4 Data Management and Preprocessing Tools

1. **NumPy and Pandas:** Core Python libraries for numerical computation and structured data handling. These tools were used for dataset organization, feature matrix preparation, and efficient batch data processing.

2. **Scikit-learn:** Used for preprocessing steps such as data normalization, splitting datasets into training and testing sets, and evaluating model performance with classification metrics.

3. **Matplotlib:** Visualization libraries used for plotting accuracy and loss curves, confusion matrices, and performance comparisons during model evaluation.

5.5 Development and Deployment Platforms

1. **Google Colab:** Cloud-based platform offering GPU and TPU support, enabling faster training and testing of deep learning models without dedicated hardware.

Chapter 6

Model Training and Implementation

This chapter focuses on the training and implementation of the machine learning models used in the proposed **Selective Noise Cancellation Using Machine Learning** system. The process involves preparing the dataset, extracting meaningful features, training the model for classification, and integrating it with the source separation pipeline to achieve real-time noise filtering.

6.1 Dataset Description

The project uses the *UrbanSound8K* dataset, which contains 8,732 labeled audio clips with a duration of up to 4 seconds each.

- The dataset consists of 10 sound classes including sirens, car horns, dog barks, drilling, jackhammers, street music, engine idling, gunshots, air conditioner sounds, and children playing.
- All audio files are in 44.1 kHz, 16-bit WAV format, ensuring consistent quality and easy preprocessing.
- The sounds are divided into critical alert sounds (important) and background noises (non-important).
- The dataset is split into training, validation, and testing subsets to evaluate model performance systematically.

6.2 Feature Extraction

To train the classification model effectively, relevant features are extracted from the audio signals. The following steps are followed:

1. **Audio Normalization:** Ensures a consistent volume level across all clips.
2. **Framing and Windowing:** Converts the audio into short overlapping frames for detailed analysis.
3. **MFCC (Mel-Frequency Cepstral Coefficients):** Captures perceptually relevant features of the sound.
4. **Spectral Features:** Includes pitch, energy, and zero-crossing rate to represent signal characteristics.
5. **Feature Scaling:** Standardization ensures better model convergence during training.

These extracted features serve as the input to the Convolutional Neural Network (CNN) classifier.

6.3 Model Training Process

1. **Source Separation:** The input audio is first processed through `Demucs`, which separates it into two components — main audio (foreground) and background noise.
2. **Label Assignment:** Important and non-important noise categories are labeled based on predefined classes (e.g., siren and horn as important, chatter and traffic as non-important).
3. **Model Selection:** A CNN-based architecture is chosen due to its strong performance in audio classification tasks.
4. **Training Setup:**
   - Optimizer: Adam
   - Loss Function: Categorical Crossentropy
   - Epochs: Determined based on convergence
   - Batch Size: Adjusted for efficient GPU utilization
5. **Validation:** A separate validation set is used to monitor overfitting and tune hyperparameters.
6. **Testing:** Final performance is measured using accuracy, precision, recall, and F1-score on unseen test data.

6.4 Signal Reconstruction and Integration

Once the classifier identifies the noise as important or non-important, the system performs:

- **Filtering:** Non-important sounds are suppressed from the background signal.
- **Merging:** The preserved important sounds are combined with the original foreground audio to reconstruct a clean and meaningful output signal.
- **Normalization:** The final signal is normalized to maintain quality and volume balance.
- **Real-time Output:** The system is capable of processing signals quickly, enabling its use in real-time scenarios like headphones or in-vehicle systems.

6.5 Implementation Details

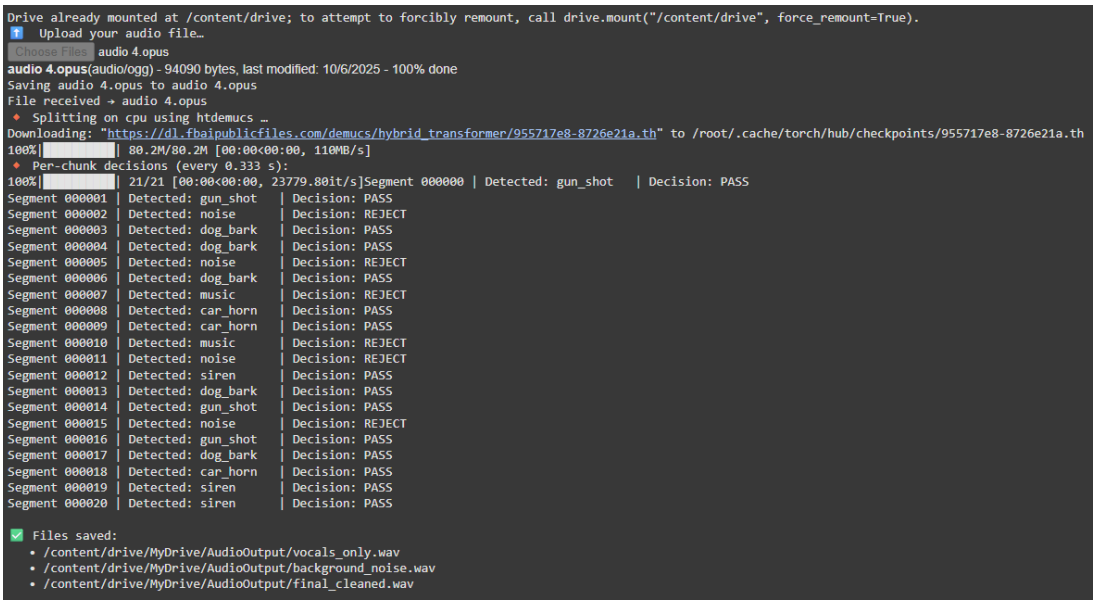
- **Programming Language:** Python
- **Frameworks:** TensorFlow and PyTorch
- **Audio Processing:** Librosa and SoundFile
- **Environment:** Google Colab with GPU acceleration for model training
- **Model Storage:** The trained model is saved and loaded for classification during runtime.
- **Integration:** The separation, classification, and reconstruction components are combined into a single processing pipeline for efficiency.

Chapter 7

Results

This chapter presents the outcomes of the proposed methodology, highlighting the implementation of the selective noise cancellation system. The results demonstrate the system's ability to separate primary audio from environmental noise and evaluate its performance on various audio samples.

7.1 Model Output – Sound Classification Results



```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
   Upload your audio file…
Choose Files  audio 4.opus
audio 4.opus(audio/ogg) - 94090 bytes, last modified: 10/6/2025 - 100% done
Saving audio 4.opus to audio 4.opus
File received → audio 4.opus
 ◆  Splitting on cpu using htdemucs …
Downloading: "https://dl.fbaipublicfiles.com/demucs/hybrid_transformer/955717e8-8726e21a.th" to /root/.cache/torch/hub/checkpoints/955717e8-8726e21a.th
100%|          | 80.2M/80.2M [00:00<00:00, 110MB/s]
 ◆  Per-chunk decisions (every 0.333 s):
100%|          | 21/21 [00:00<00:00, 23779.80it/s]Segment 000000 | Detected: gun_shot   | Decision: PASS
Segment 000001 | Detected: gun_shot   | Decision: PASS
Segment 000002 | Detected: noise      | Decision: REJECT
Segment 000003 | Detected: dog_bark   | Decision: PASS
Segment 000004 | Detected: dog_bark   | Decision: PASS
Segment 000005 | Detected: noise      | Decision: REJECT
Segment 000006 | Detected: dog_bark   | Decision: PASS
Segment 000007 | Detected: music      | Decision: REJECT
Segment 000008 | Detected: car_horn   | Decision: PASS
Segment 000009 | Detected: car_horn   | Decision: PASS
Segment 000010 | Detected: music      | Decision: REJECT
Segment 000011 | Detected: noise      | Decision: REJECT
Segment 000012 | Detected: siren      | Decision: PASS
Segment 000013 | Detected: dog_bark   | Decision: PASS
Segment 000014 | Detected: gun_shot   | Decision: PASS
Segment 000015 | Detected: noise      | Decision: REJECT
Segment 000016 | Detected: gun_shot   | Decision: PASS
Segment 000017 | Detected: dog_bark   | Decision: PASS
Segment 000018 | Detected: car_horn   | Decision: PASS
Segment 000019 | Detected: siren      | Decision: PASS
Segment 000020 | Detected: siren      | Decision: PASS

 ✅  Files saved:
   • /content/drive/MyDrive/AudioOutput/vocals_only.wav
   • /content/drive/MyDrive/AudioOutput/background_noise.wav
   • /content/drive/MyDrive/AudioOutput/final_cleaned.wav
```

Figure 7.1: Model output showing segment-wise sound detection and classification results

The trained CNN model analyzes the audio file in short segments to detect sound types. Each segment is marked as PASS or REJECT based on detection confidence. The system then produces separate outputs - vocals only, background noise, and a final cleaned audio file — showcasing effective sound classification and noise separation for urban sound analysis.

7.2 Confusion Matrix of UrbanSound CNN Model

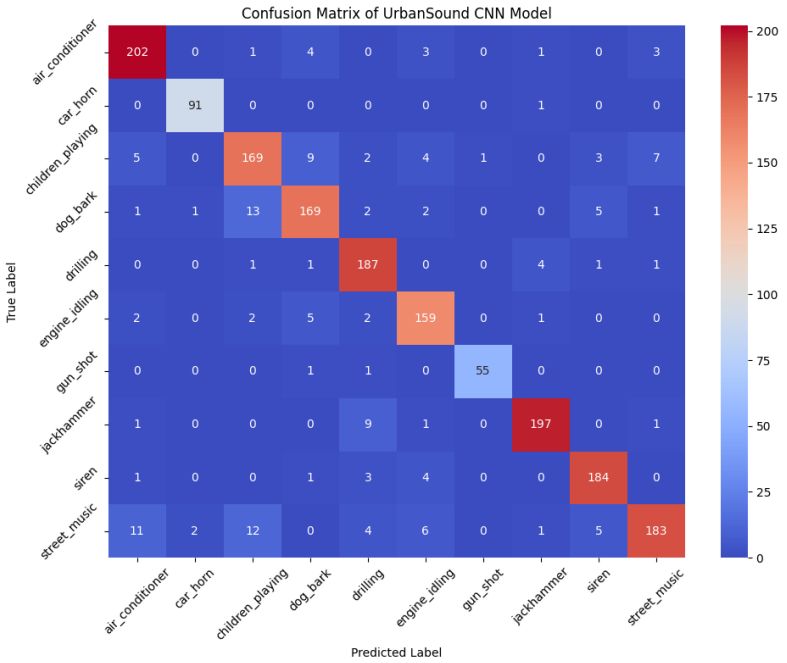


Figure 7.2: Confusion matrix showing CNN model performance across 10 sound classes.

7.3 Training and Validation Curves

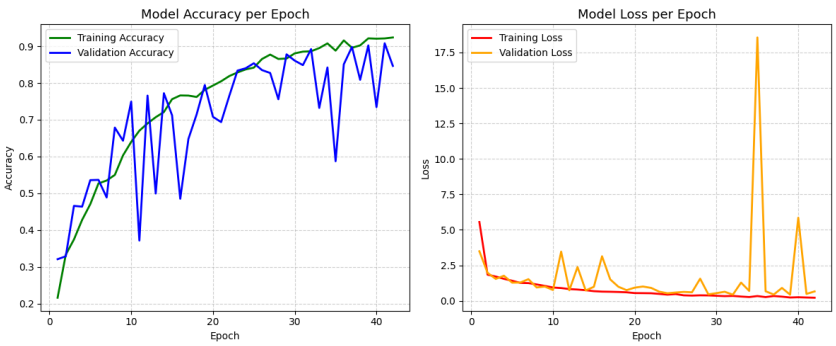


Figure 7.3: Training and validation curves showing the CNN model's accuracy and loss over 42 epochs.

Chapter 8

Conclusion and Future Work

This chapter presents the overall conclusion of the mini project **"Selective Noise Cancellation Using Machine Learning"** and highlights potential future improvements. The system was developed with the goal of intelligently identifying and preserving essential sounds (such as sirens and alarms) while suppressing irrelevant background noise in real time.

8.1 Summary of Achievements

The proposed system successfully integrates machine learning and audio processing techniques to achieve selective noise cancellation. Key accomplishments include:

- Implementation of a **source separation technique** using a Machine learning model to isolate foreground and background audio components.
- Development of a **feature extraction pipeline** using MFCCs, spectral features, and amplitude-related parameters to capture relevant sound characteristics.
- **Training and deployment of a CNN classifier** to differentiate between important (e.g., sirens, horns) and non-important (e.g., chatter, drilling, engine noise) sounds.

This confirms that machine learning can effectively enhance traditional noise cancellation systems by adding context-awareness and selectivity.

8.2 Potential Improvements

Although the project achieved its core objectives, there are several areas where further improvements can be made:

- **Improved Sound Classification:** More advanced deep learning architectures or transformer-based audio models could further improve classification accuracy, especially in noisy and overlapping environments.

- **Support for More Sound Categories:** The current model focuses on a limited number of sound classes. Expanding the dataset to include more diverse and context-specific sounds would make the system more robust.
- **Latency Reduction:** Optimizing the processing pipeline would enable faster and more efficient real-time performance, particularly for use in embedded systems.
- **Noise Adaptation:** Dynamic noise profiling and adaptive filtering techniques can be explored to make the system more responsive to changing environments.

8.3 Future Directions

The project can be extended in multiple directions to increase its real-world applicability:

- **Integration with Wearable and Automotive Systems:** Embedding the model into wireless headphones, hearing aids, or vehicle infotainment systems can enhance user safety and comfort.
- **Edge Deployment and Optimization:** Model compression, quantization, and pruning can enable efficient deployment on low-power hardware without compromising accuracy.
- **Multimodal Sound Analysis:** Combining audio processing with other modalities (e.g., camera or sensor data) could enhance accuracy in complex environments.
- **Adaptive Learning:** Implementing continuous or online learning capabilities would allow the system to adapt to new noise patterns without complete retraining.
- **Enhanced Alert Detection:** Advanced alert detection systems can be integrated to recognize more critical events and improve user awareness in real-world scenarios.

8.4 Conclusion

In conclusion, this mini project demonstrates that **machine learning can make noise cancellation systems smarter and safer** by distinguishing between important and non-important sounds. The use of source separation and classification enables selective filtering, making the system suitable for real-time applications.

With further improvements in accuracy, latency, and adaptability, this system has the potential to be integrated into **everyday audio devices** to enhance safety, awareness, and listening comfort in noisy environments.

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